

# Ali Haider

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■ Master's in Engineering Sciences – Applied Mathematics (2024)

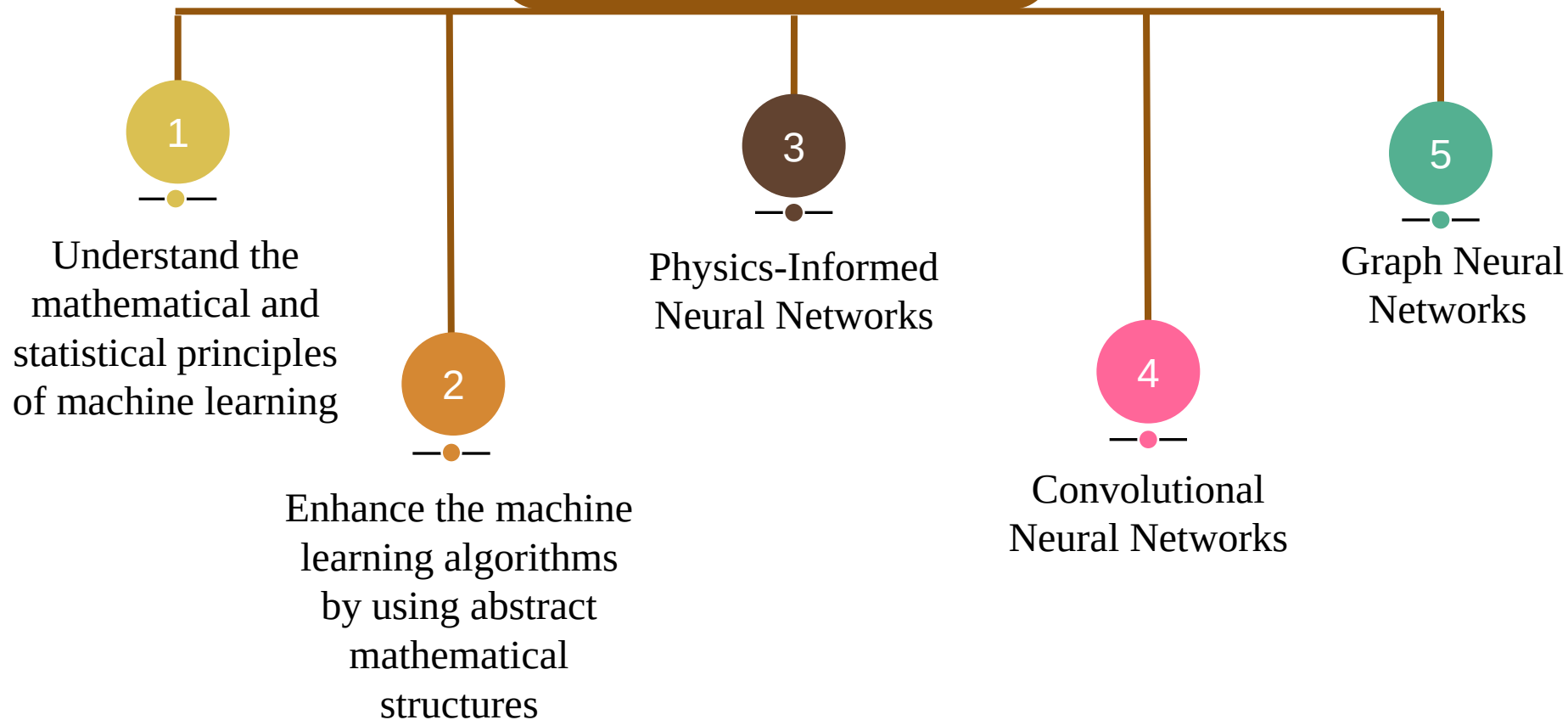
■ Faculty of Engineering Sciences

■ Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)





## RESEARCH AREAS







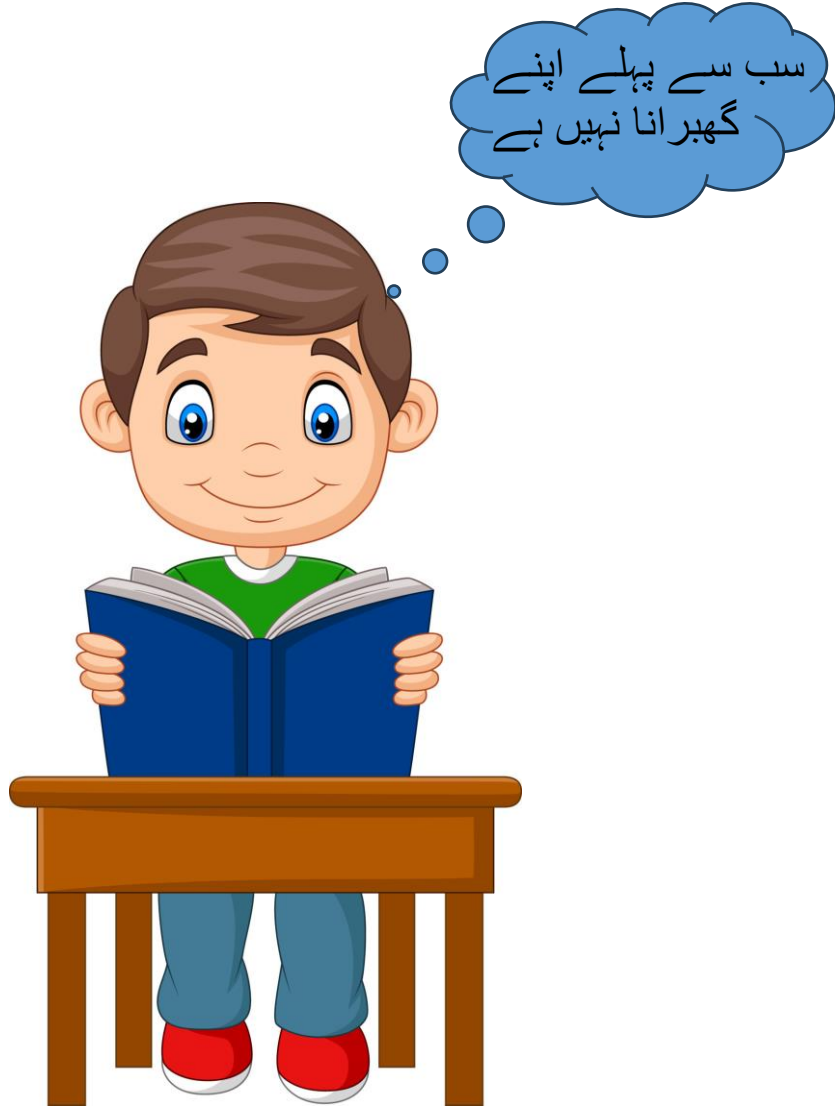
# Travelling

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Me

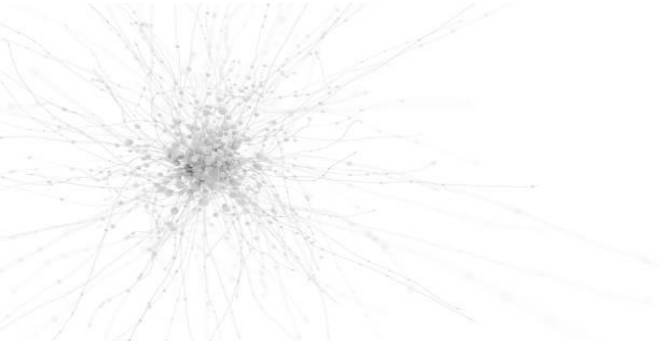


# Literature





# Introduction of Course

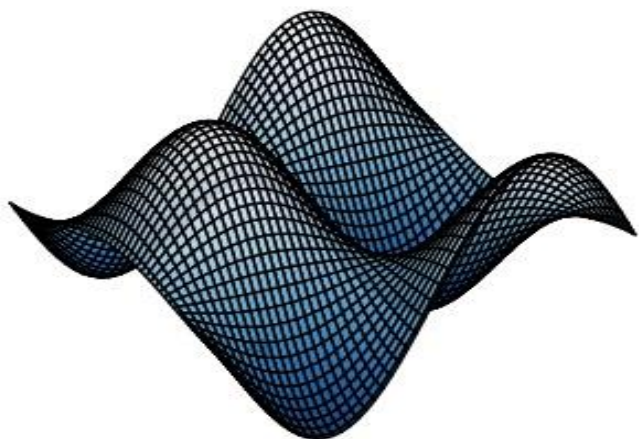


## Introduction

# Multivariable Calculus (MTH 111)

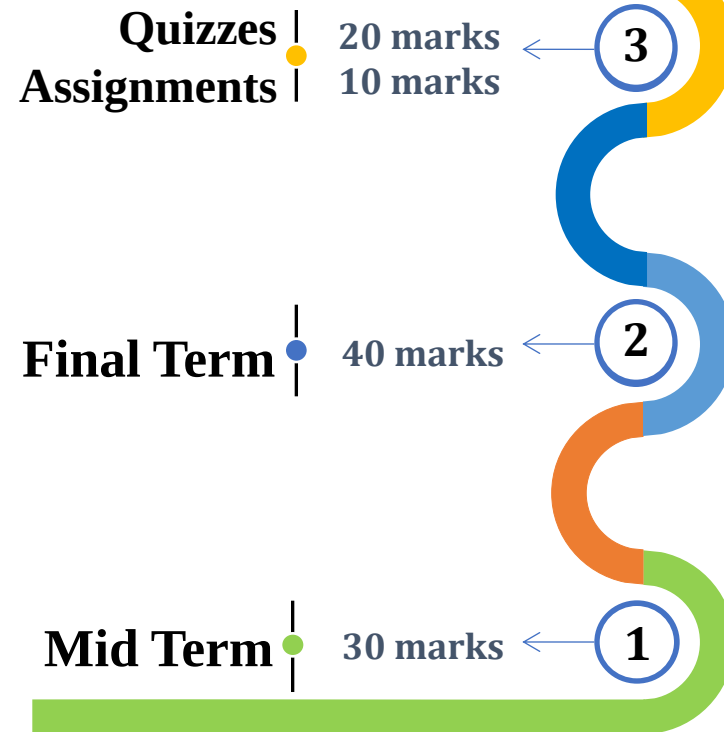
**Pre-requisite: Calculus & Analytical  
Geometry**

**Fall 2024  
Credit Hours: 3**



## Marks Distribution

Quiz 3: 5 marks  
Assignment 3: 2.5 marks



Quiz 1: 5 marks  
Assignment 1: 2.5 marks

Quiz 4: 5 marks  
Assignment 4: 2.5 marks

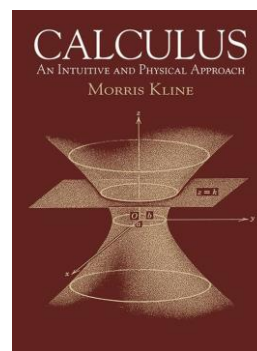
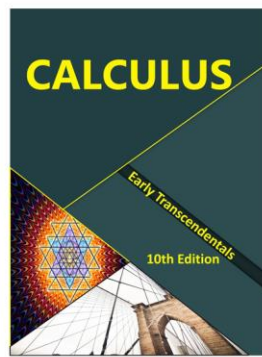
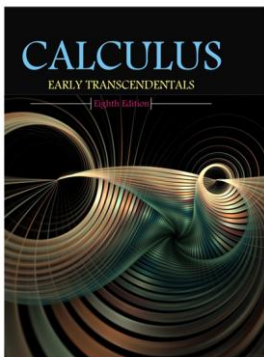
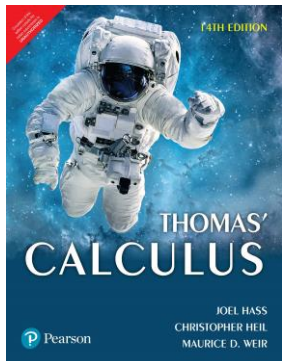
Quiz 2: 5 marks  
Assignment 2: 2.5 marks



## Books

# Thomas Calculus

By: **George B. Thomas**  
**13<sup>th</sup> Edition**



### Reference Books

**Calculus: Early  
Transcendentals**  
By: James Stewart

**Calculus**  
By: Swokowski

**Calculus**  
By: Howard Anton

# Lecture - 1



## Parametric Curves

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**Students should be familiar with:**

- **Relation?**
- **Function?**
- **Curve and Path**

## Function and Relation

**Definition of Relation** A relation between two sets is a collection of ordered pairs, where each pair consists of an element from one set (called the domain) and an element from another set (called the codomain). A relation doesn't necessarily have to map every element from the domain to a unique element in the codomain.

**Example:** Let  $A=\{1,2,3\}$  and  $B=\{4,5,6\}$ . A relation  $R$  from  $A$  to  $B$  can be:  $R=\{(1,4),(2,5),(3,6)\}$ .

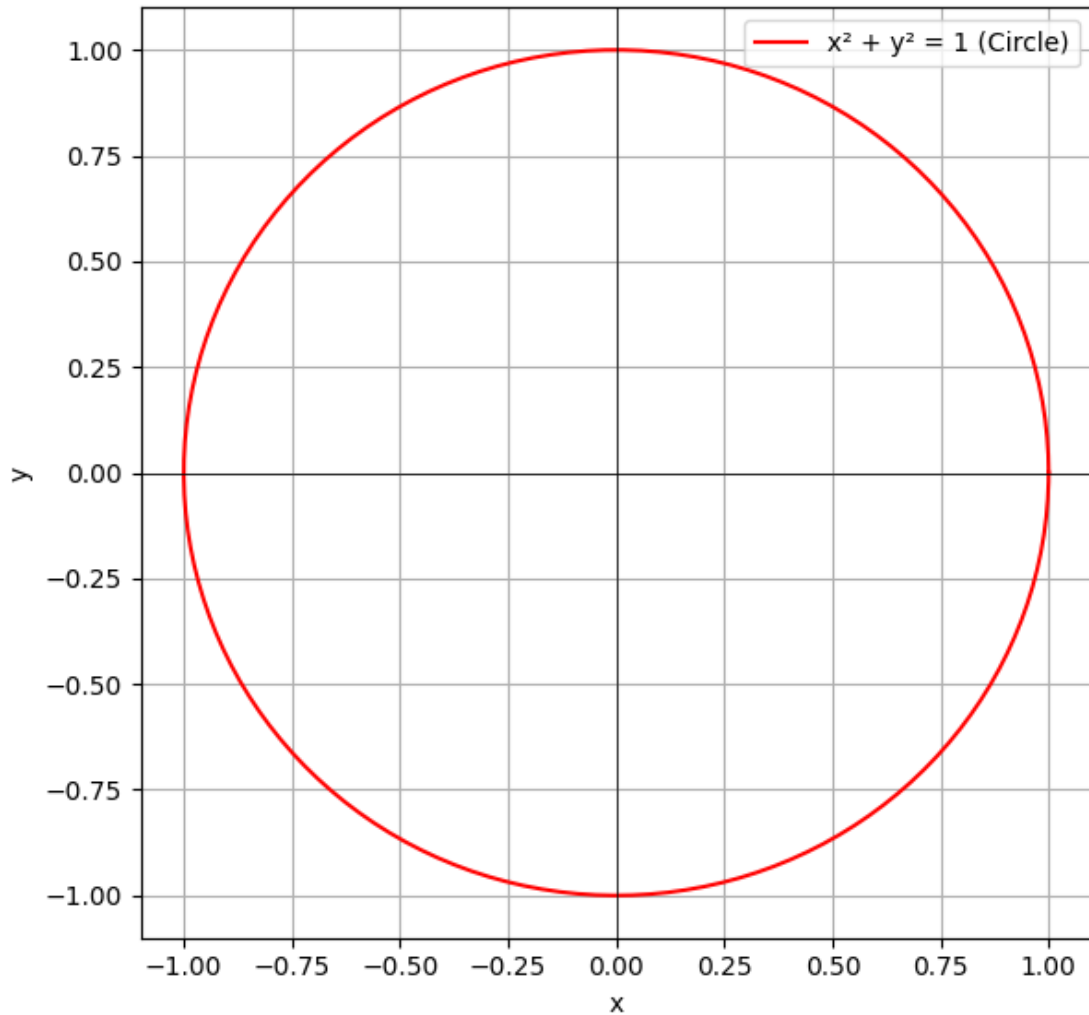
**Definition of Function** A function is a special type of relation where every element in the domain is mapped to exactly one element in the codomain. This ensures that for each input, there is only one output.

**Example:** Consider the function  $f:A\rightarrow B$ , where  $A=\{1,2,3\}$  and  $B=\{4,5,6\}$ . A possible function  $f$  is:  $f(1)=4, f(2)=5, f(3)=6$ .

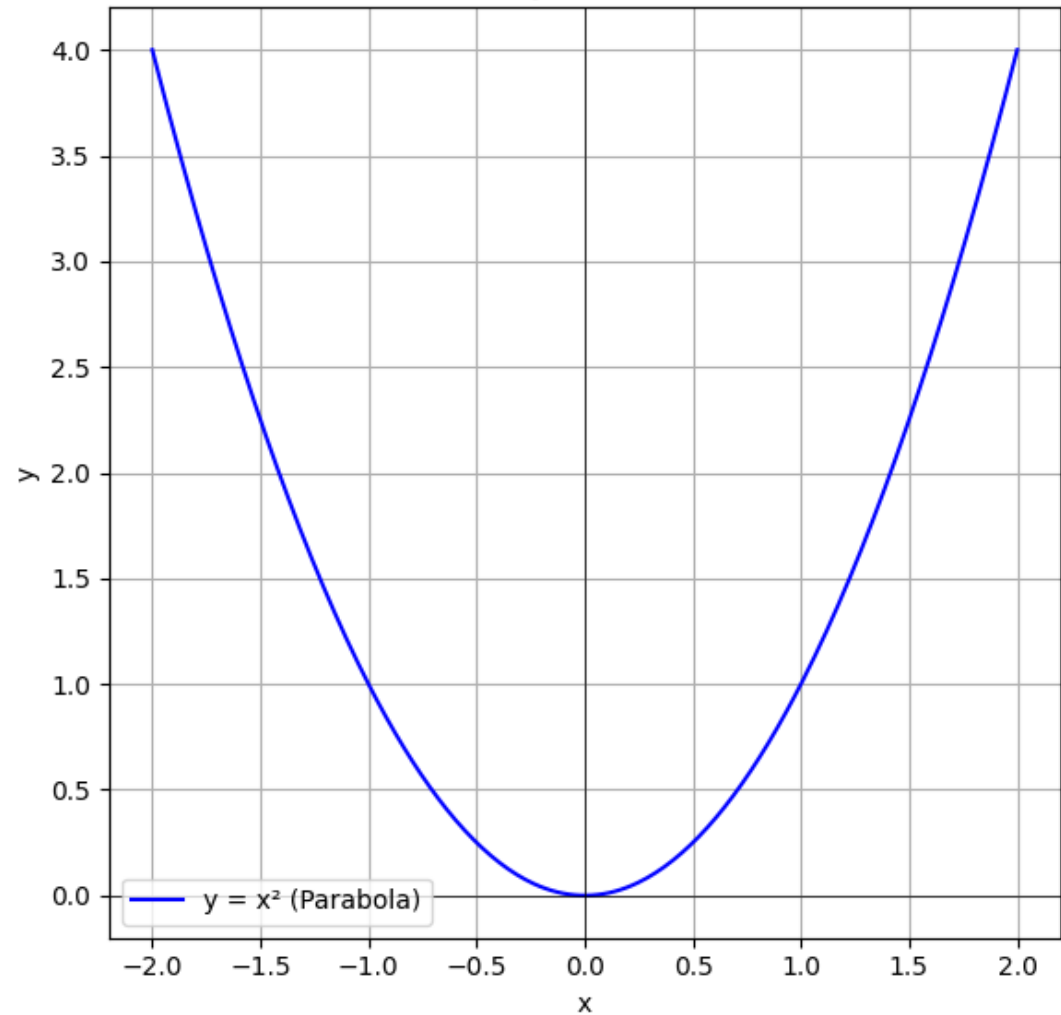


# Function & Relation (Not Function)

Relation (Not a Function): Circle

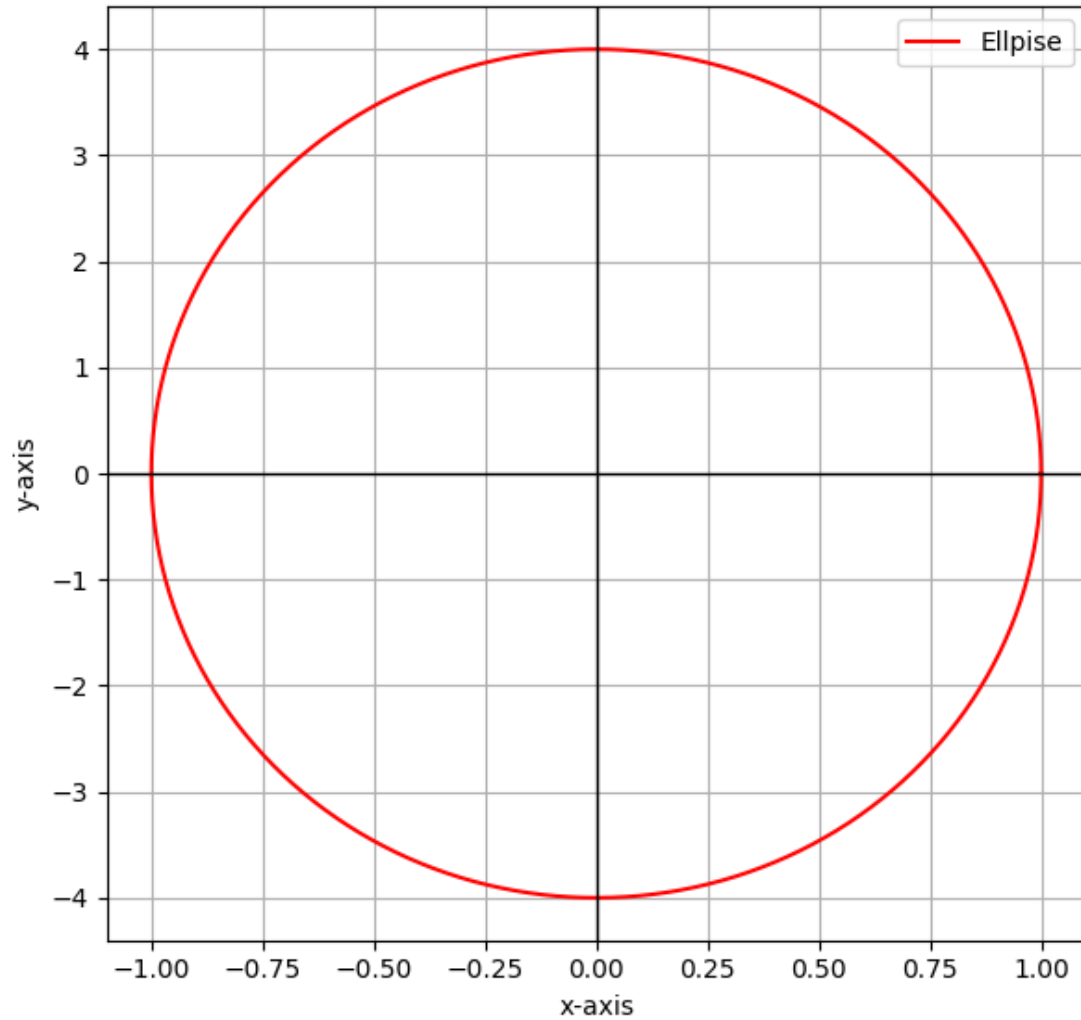


Function:  $y = x^2$  (Parabola)

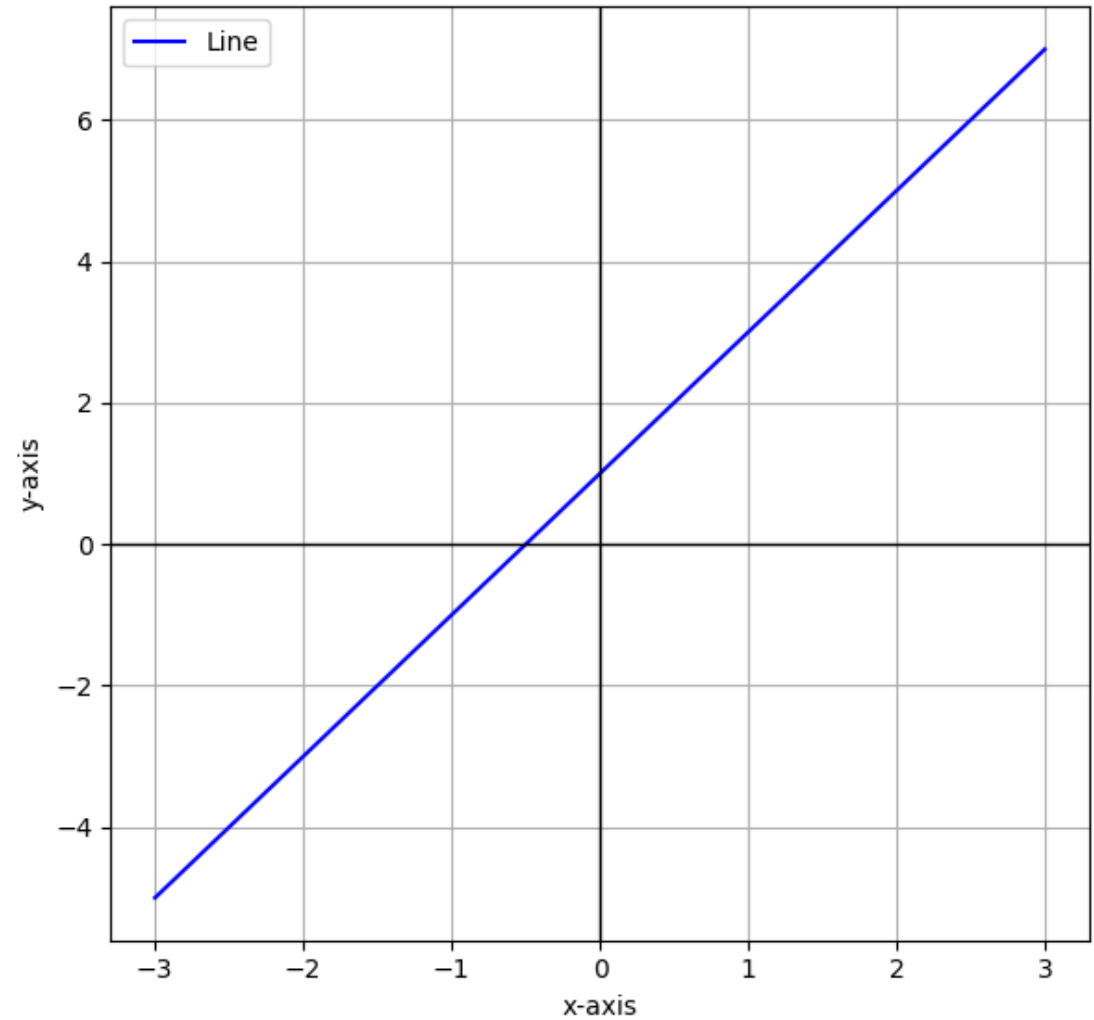


# Function & Relation (Not Function)

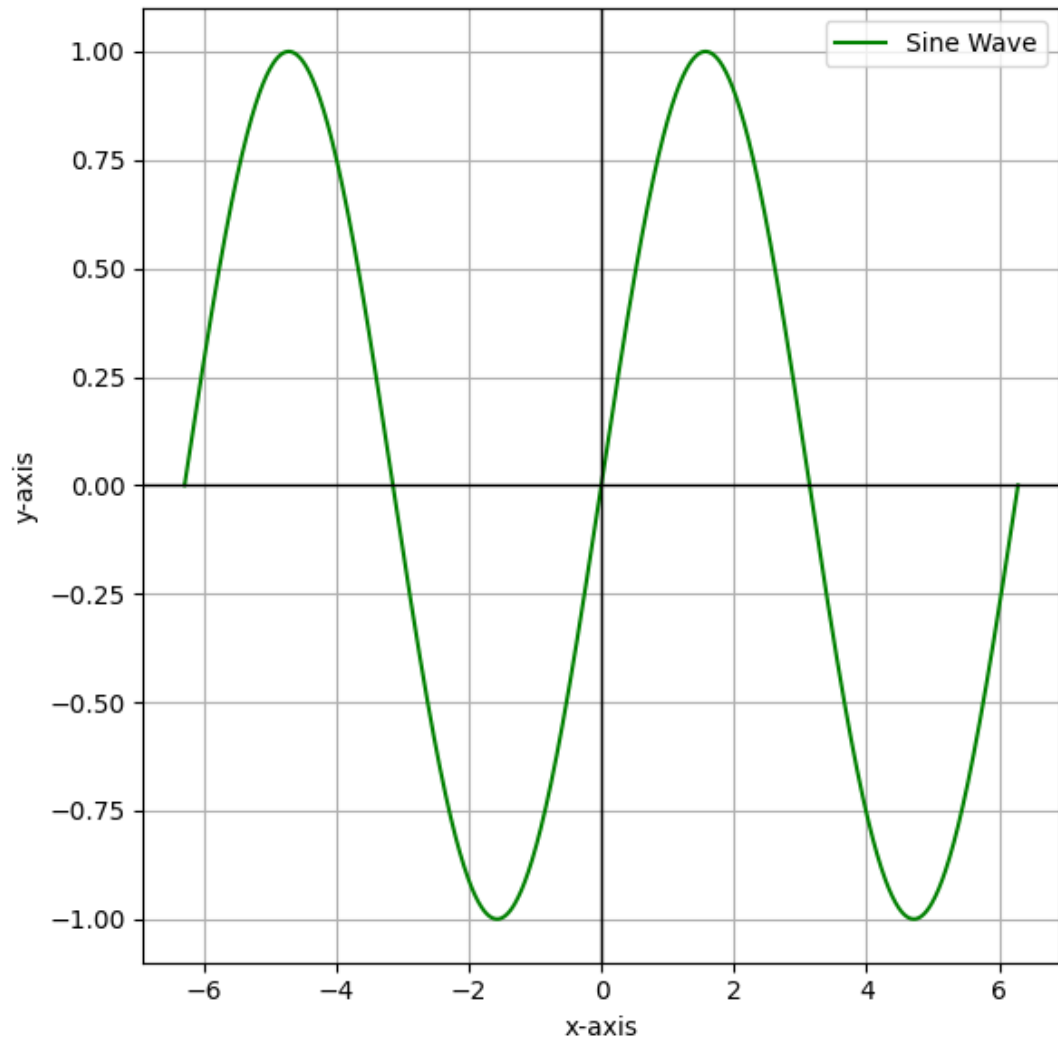
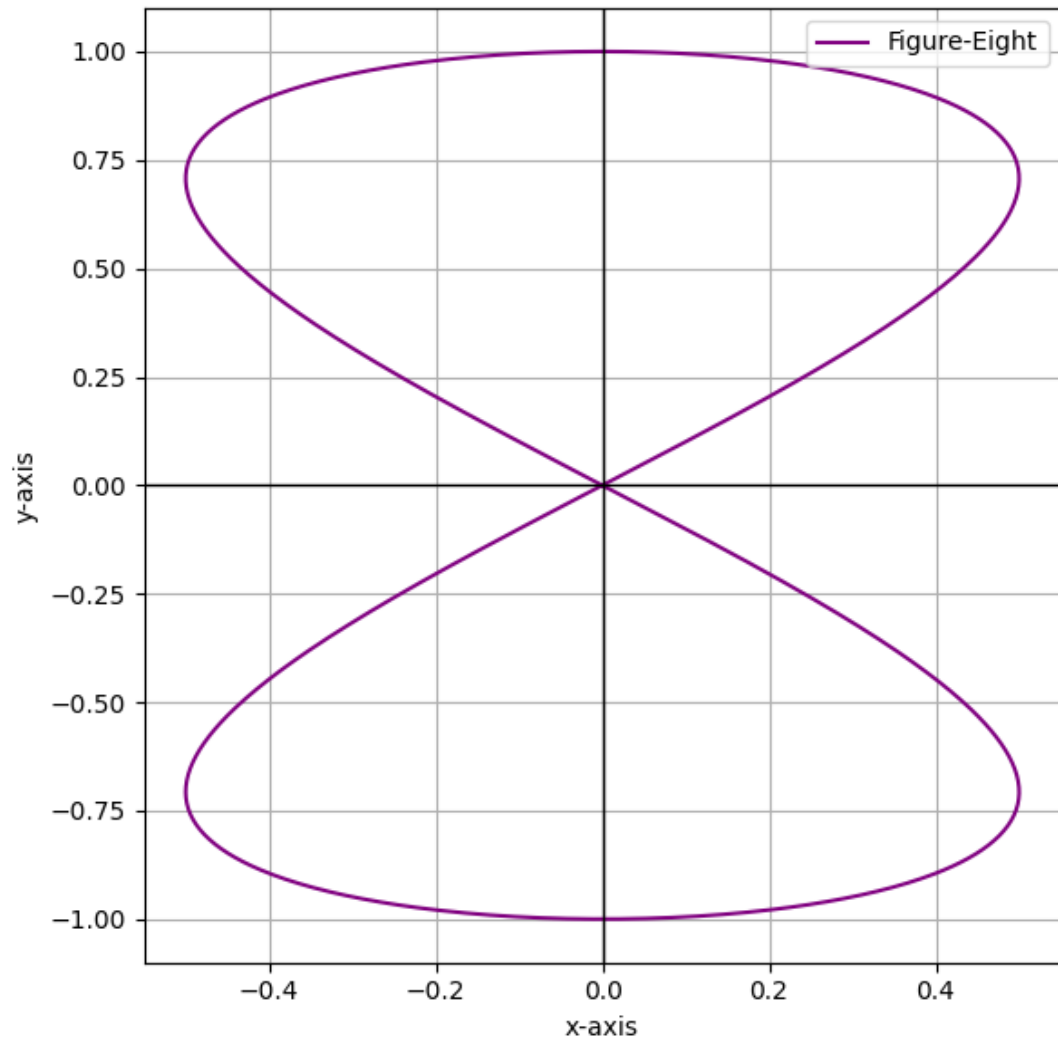
Ellipse



Line

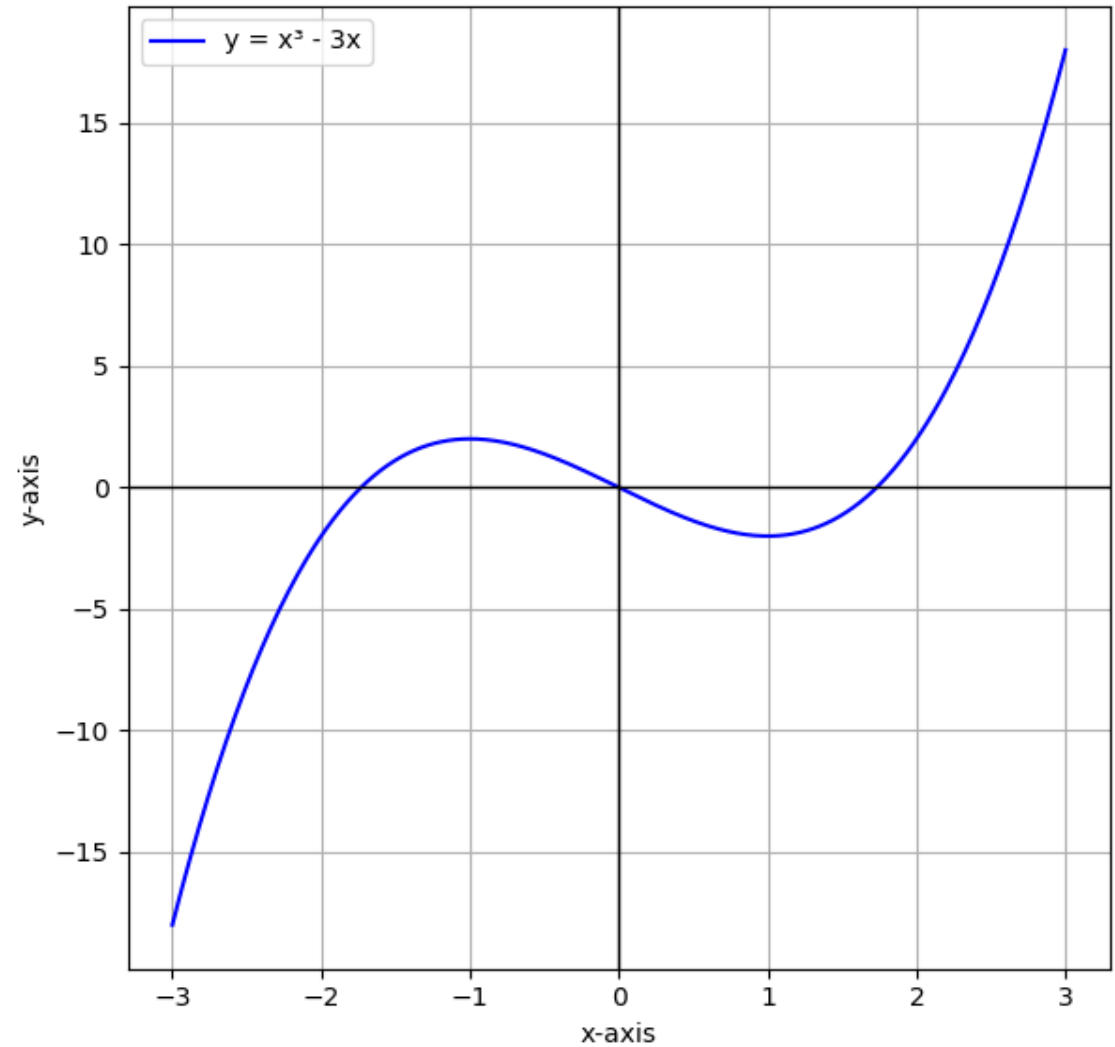
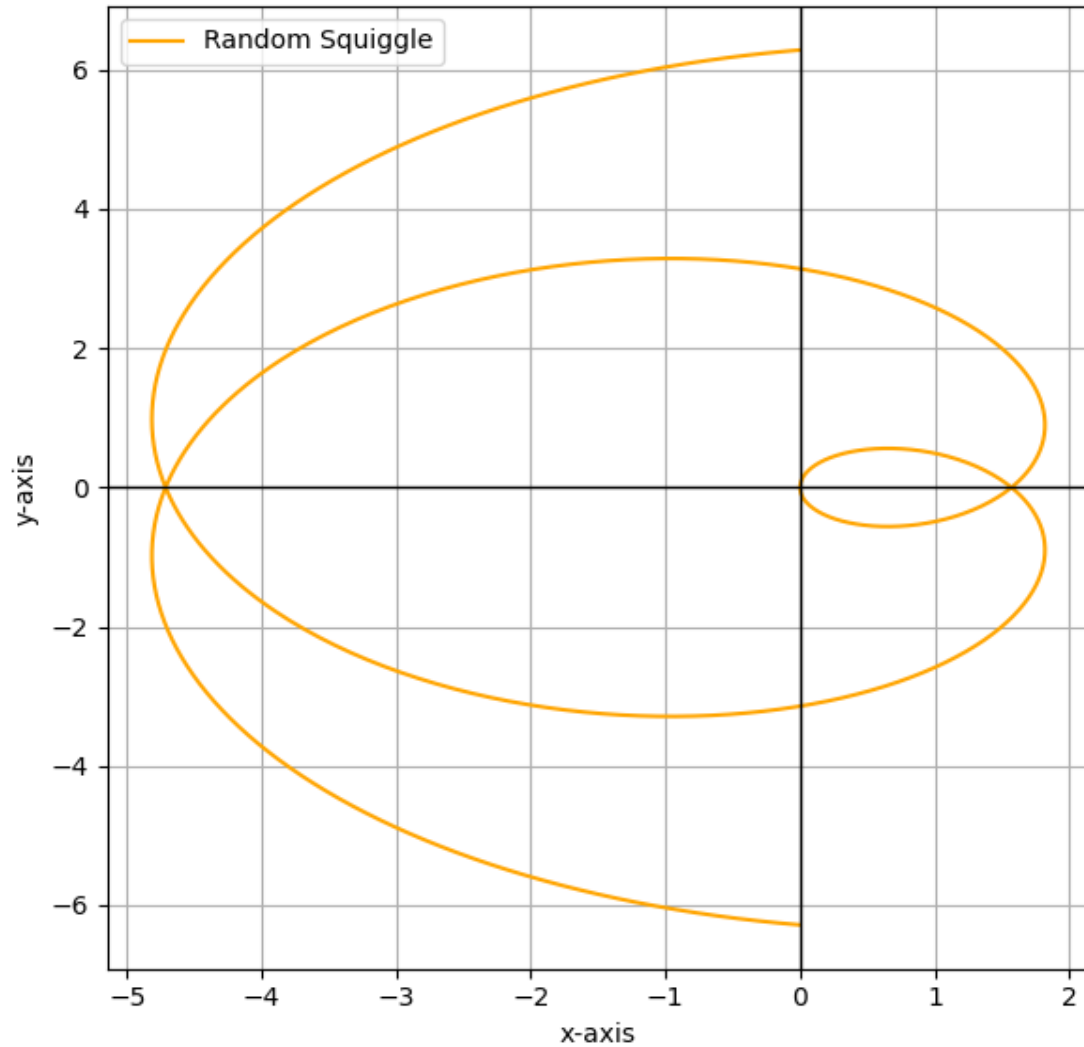


## Function & Relation (Not Function)





## Function & Relation (Not Function)

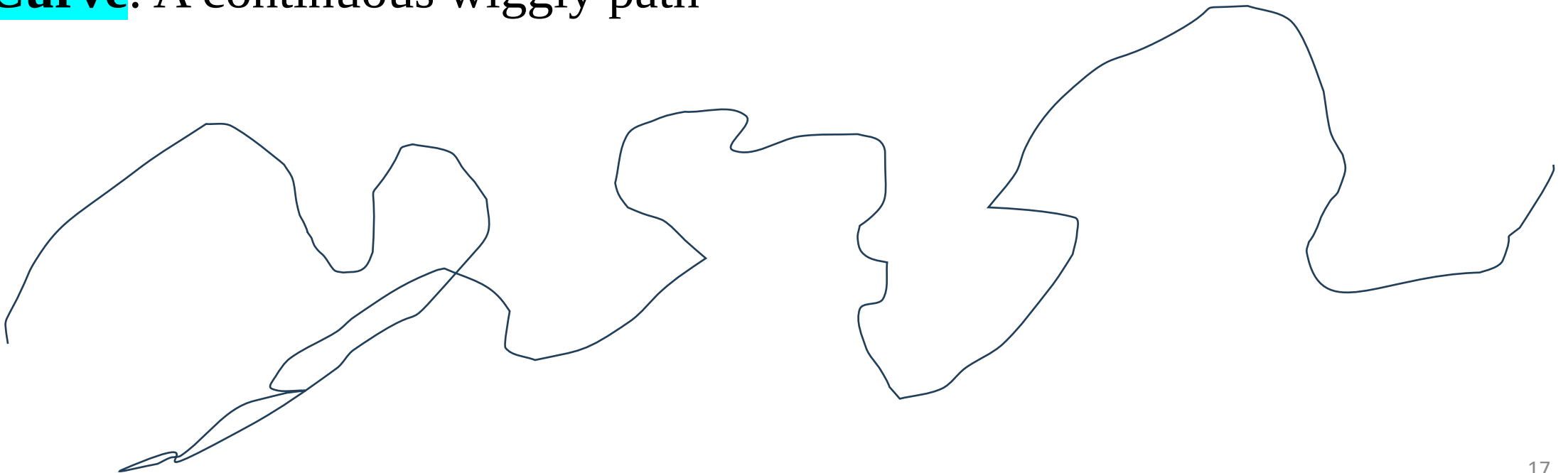


## Curve and Path

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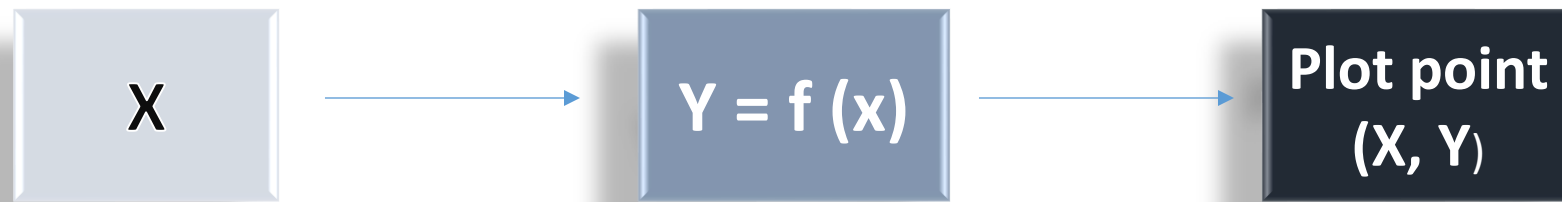
**Curve:** A curve is an object similar to line, but does not have to be straight

**Curve:** A continuous wiggly path

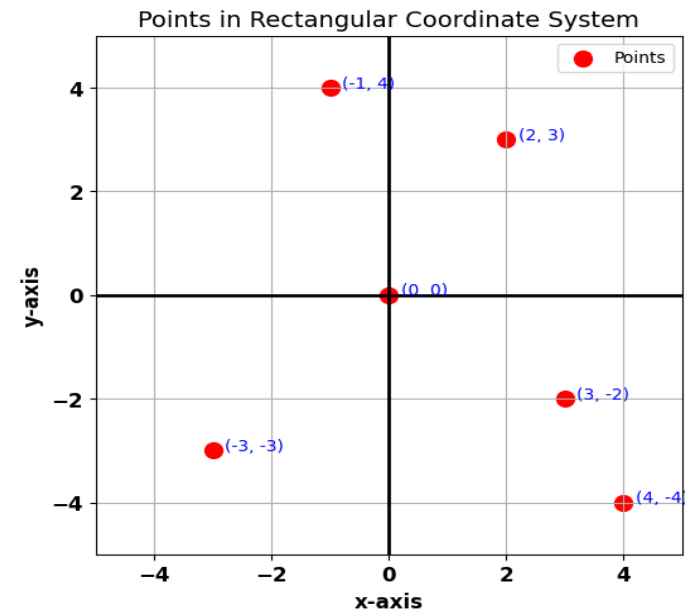


## Sketching Shapes

We are familiar with sketching shapes, such as lines, parabolas, by following basic procedure:



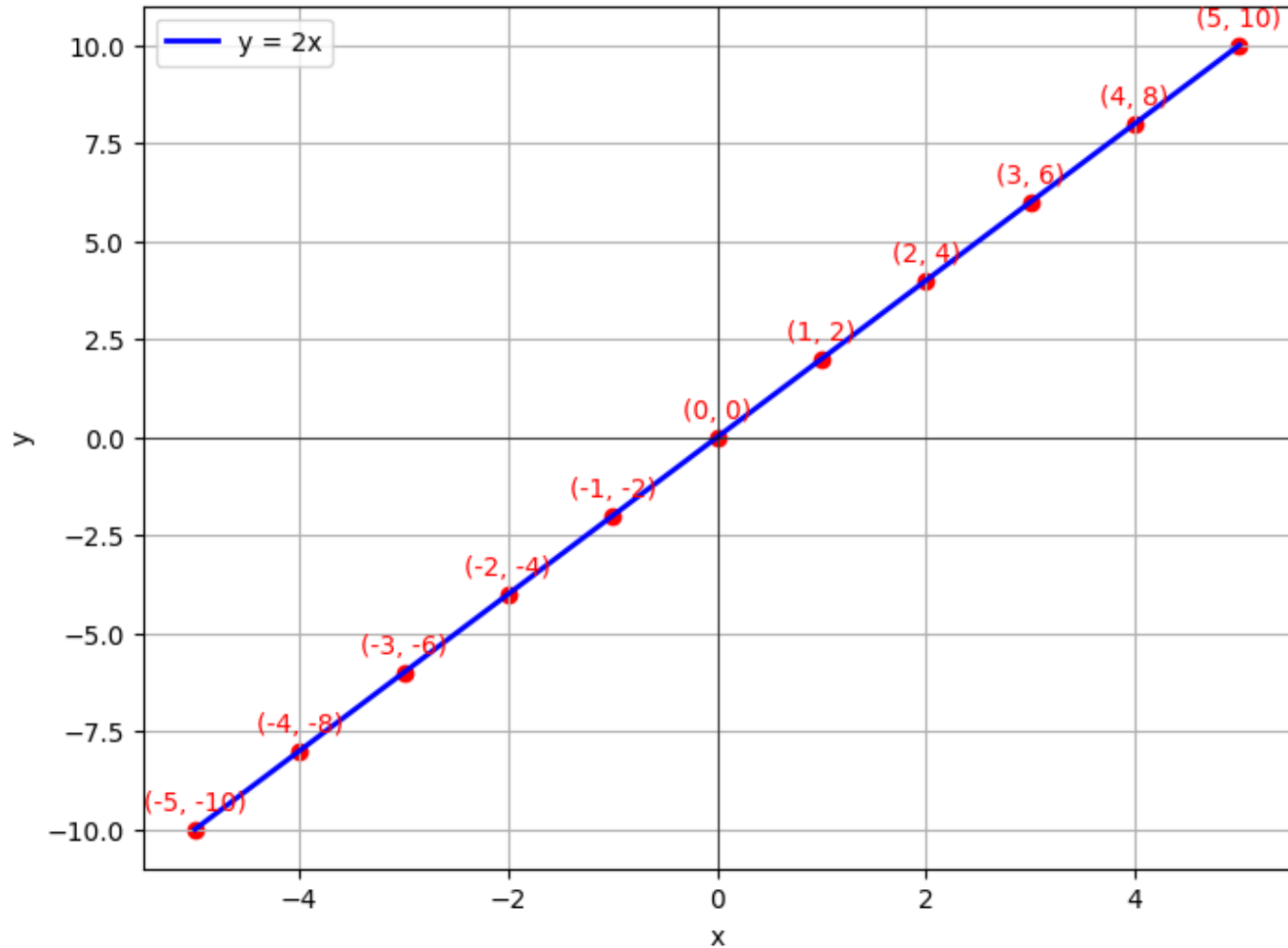
Plot the points i.e., (0,0), (2,3), (-1,4), (3, -2), (-3,-3), (4,-4) in rectangular coordinated system





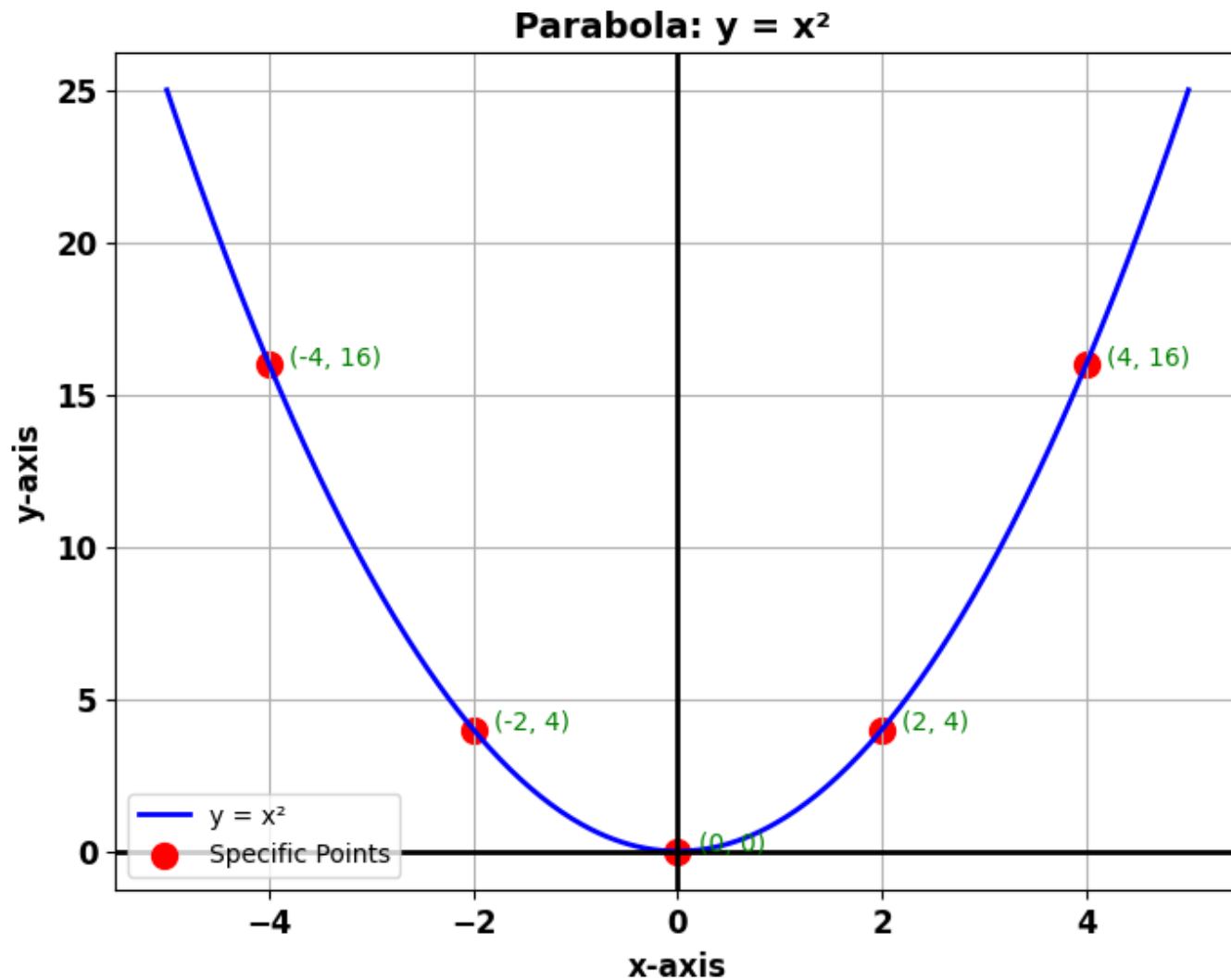
# Linear Function: $Y = 2X$

Plot of  $y = 2x$  from  $x = -5$  to  $x = 5$



| X  | Y   |
|----|-----|
| -5 | -10 |
| -4 | -8  |
| -3 | -6  |
| -2 | -4  |
| -1 | -2  |
| 0  | 0   |
| 1  | 2   |
| 2  | 4   |
| 3  | 6   |

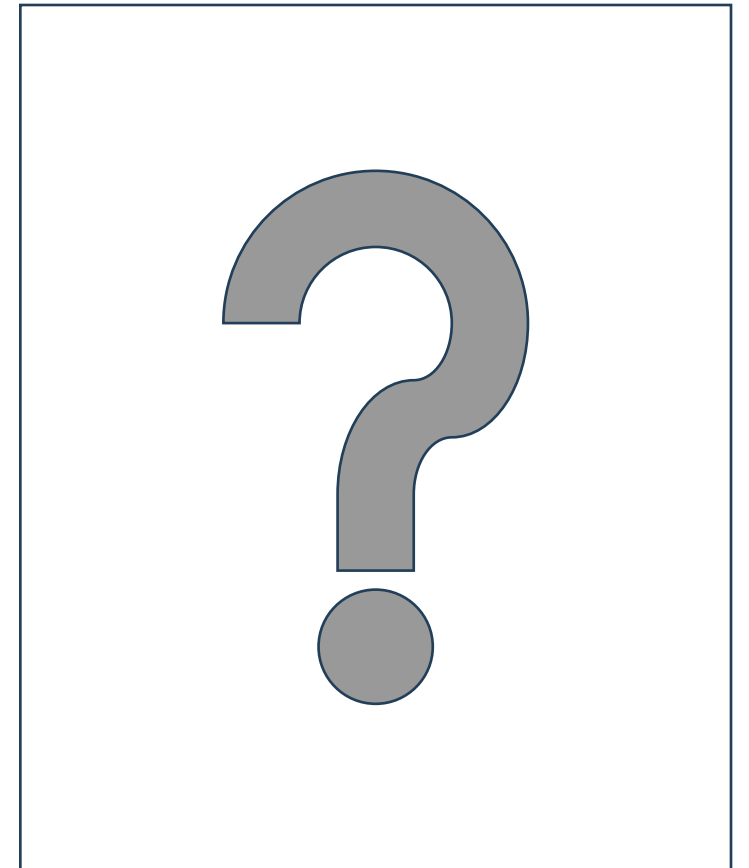
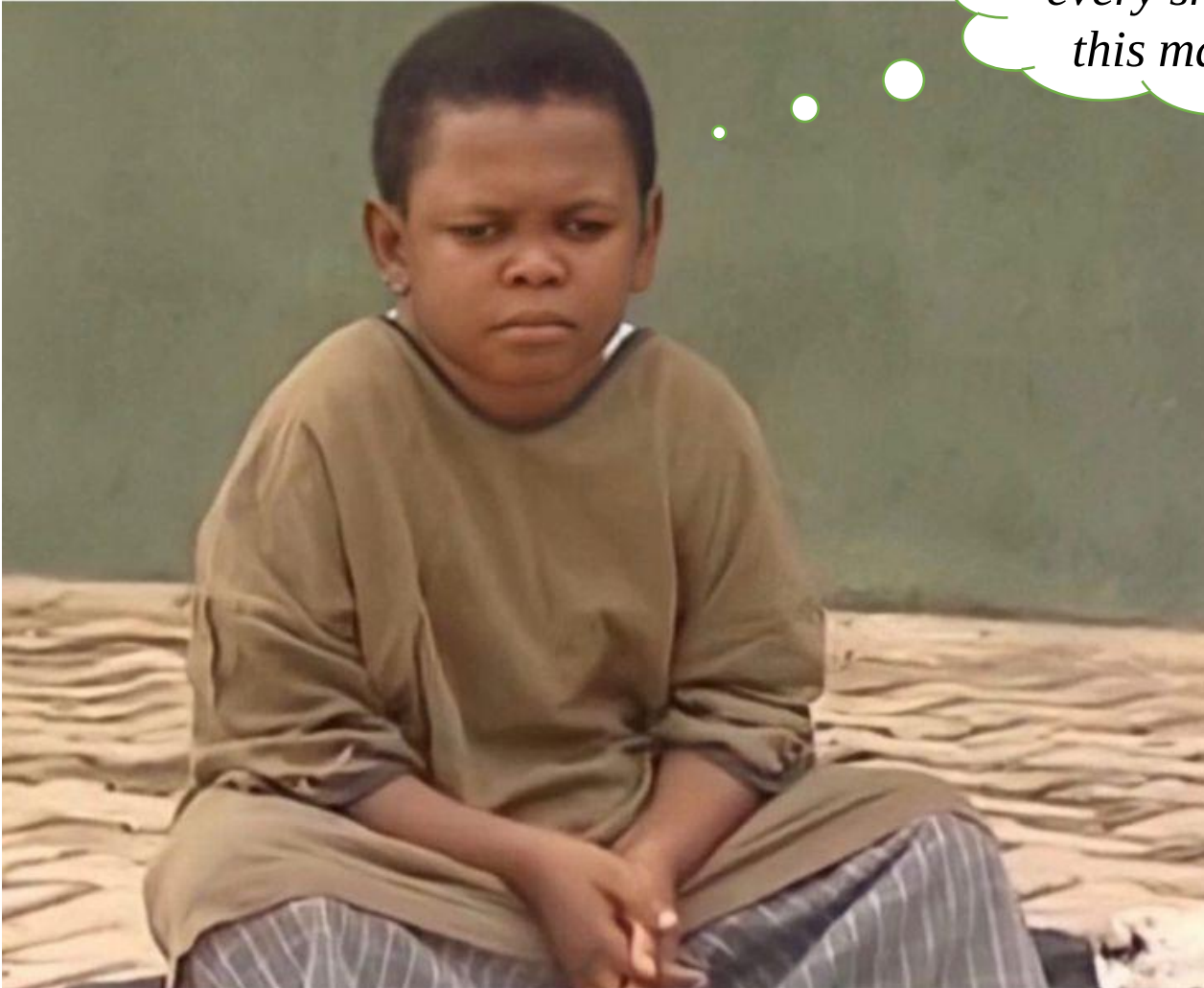
## Parabola: $Y = X^2$



| X  | Y  |
|----|----|
| -5 | 25 |
| -4 | 16 |
| -3 | 9  |
| -2 | 4  |
| -1 | 1  |
| 0  | 0  |
| 1  | 1  |
| 2  | 4  |
| 3  | 6  |

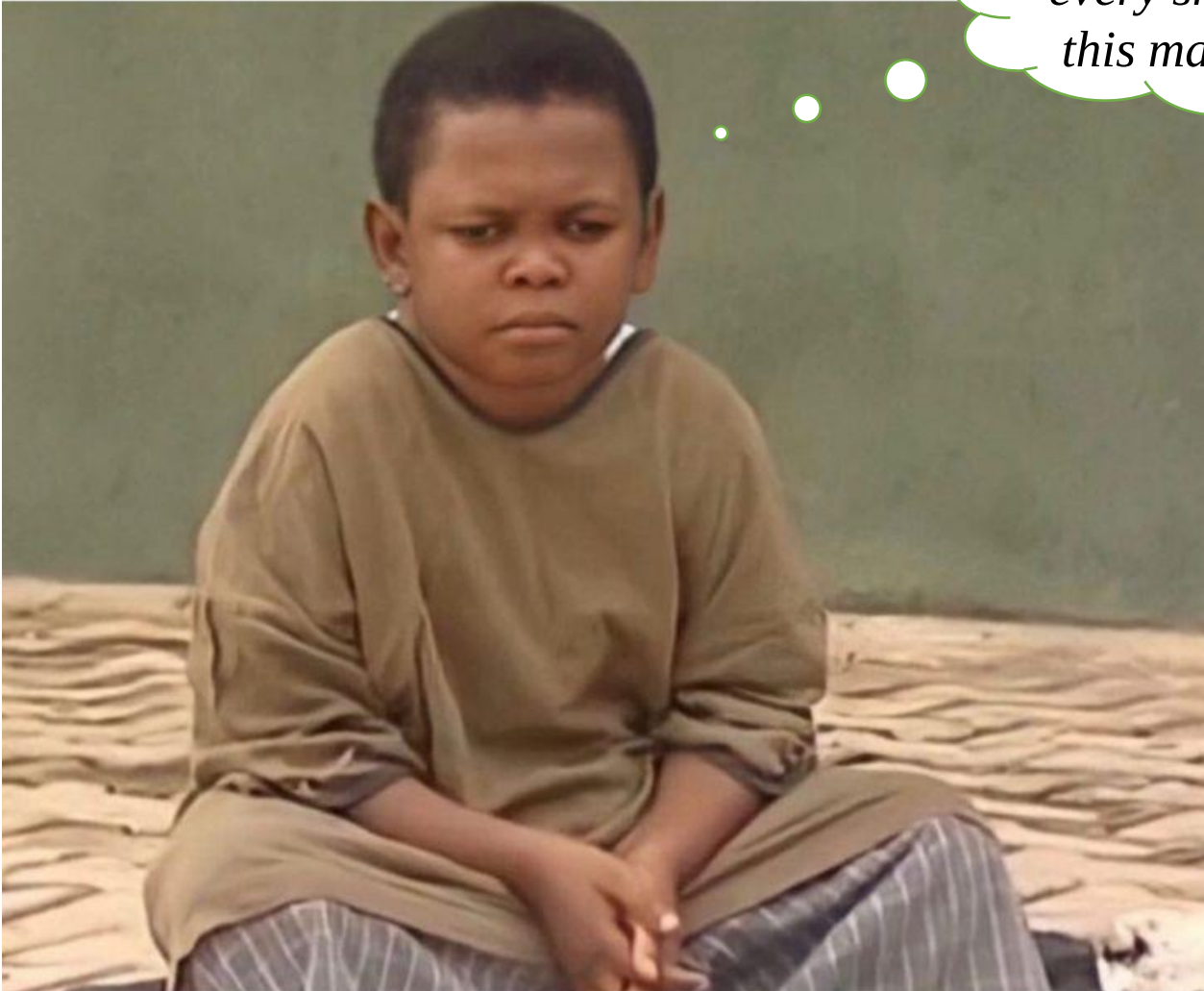
## Question

*Can we plot  
every shape in  
this manner*



## Question

*Can we plot  
every shape in  
this manner?*

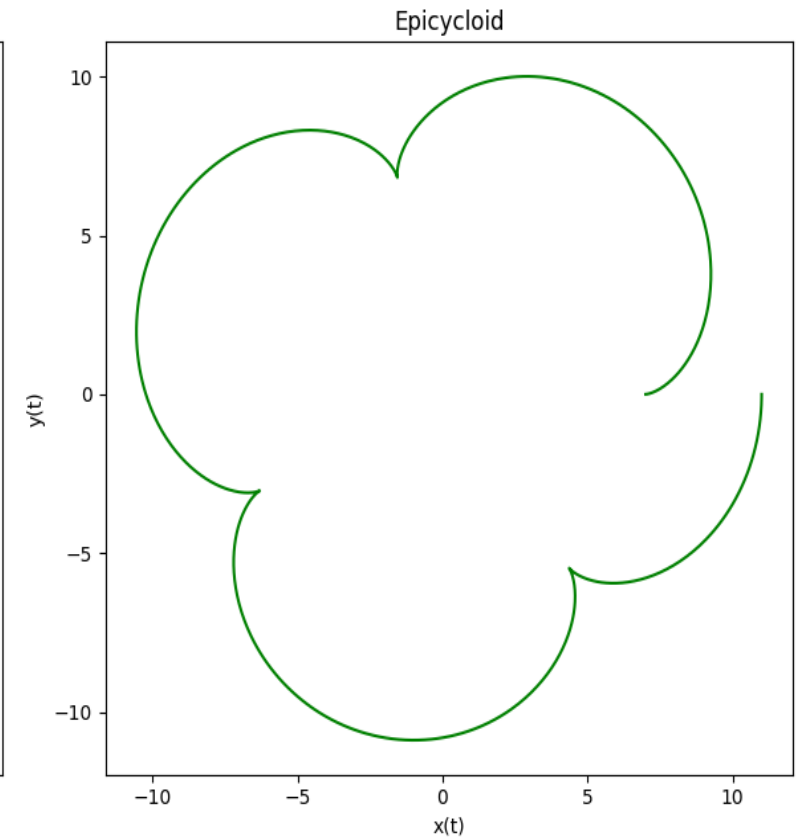
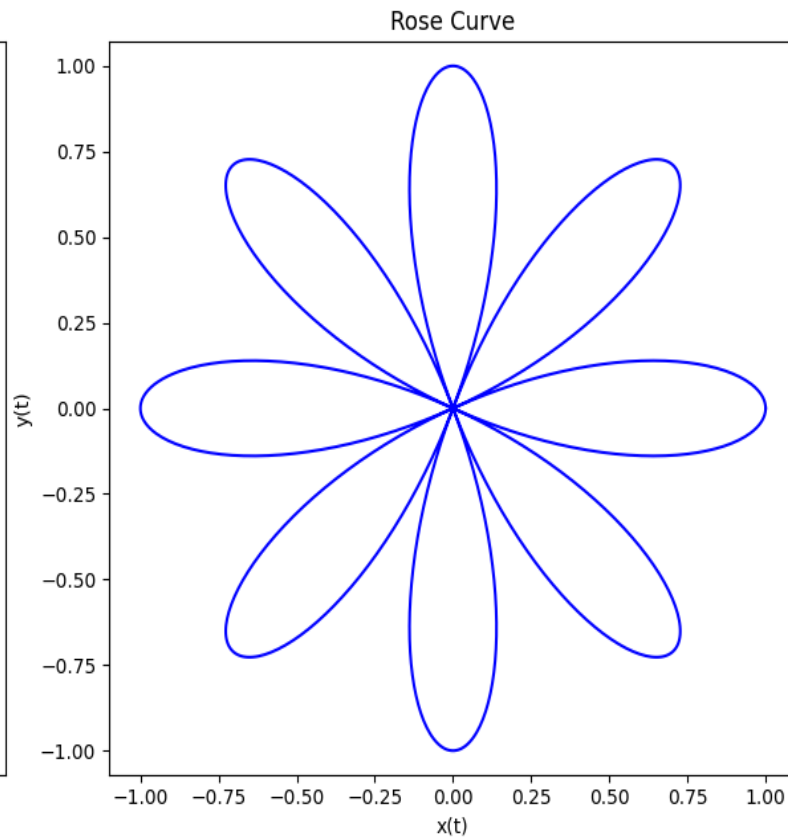
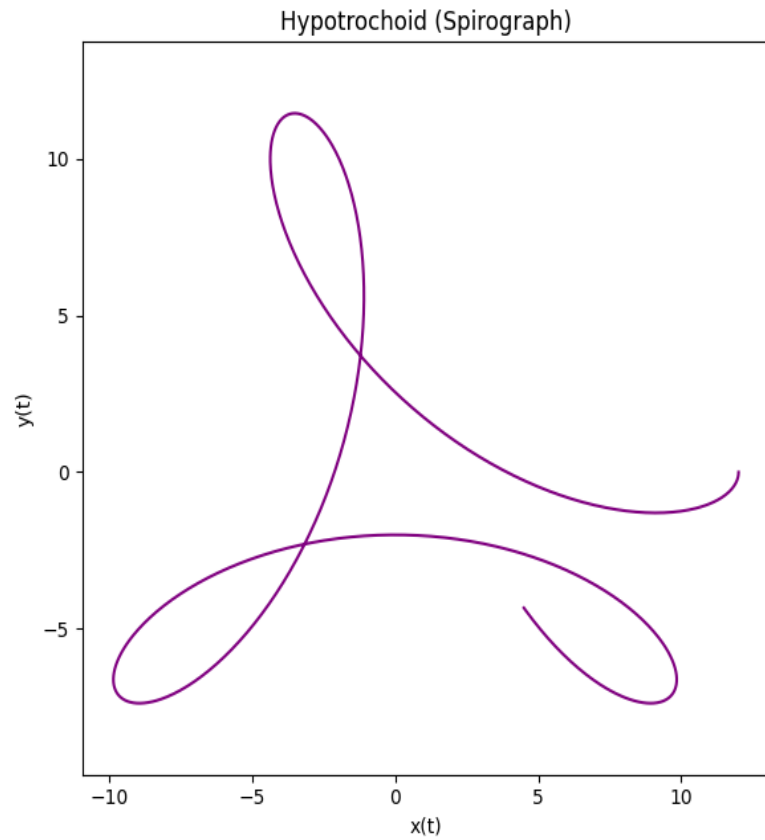


# NO

There are several relations and functions which are difficult or impossible to write in form of  **$Y = f(X)$**



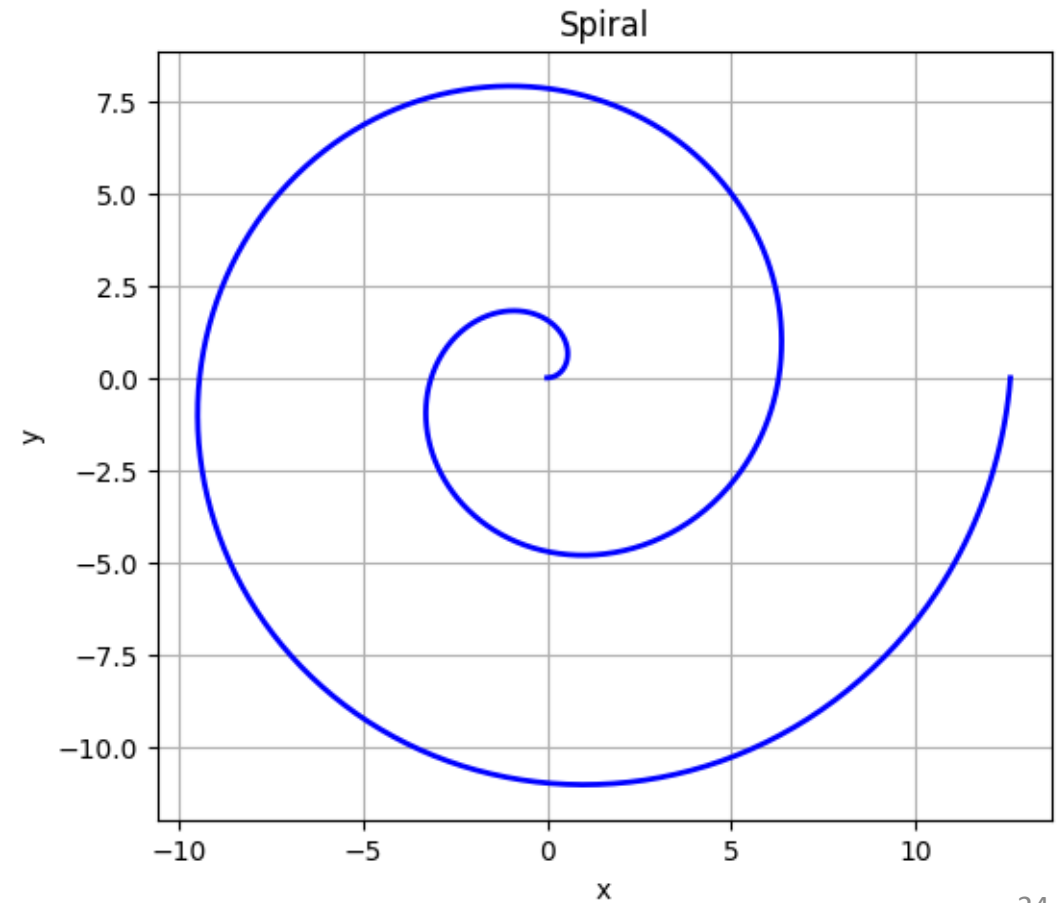
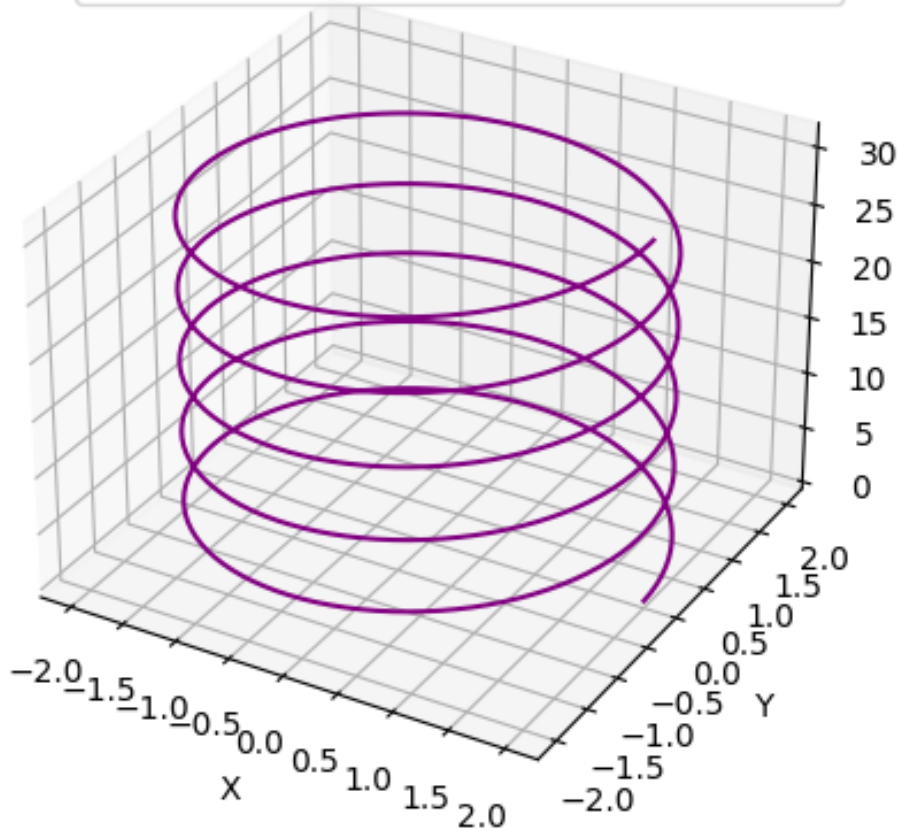
## Some Important Relations



## Some Important Relations

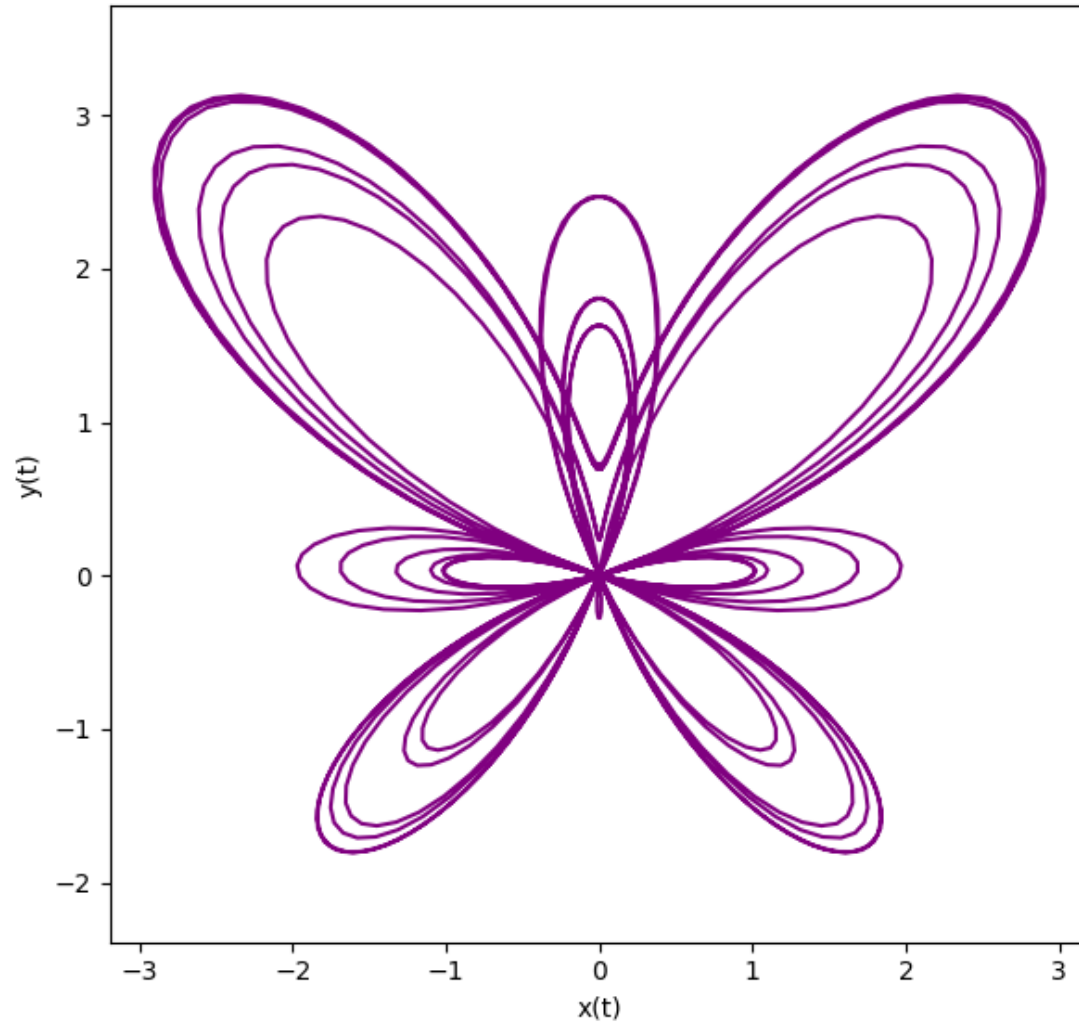
### Parametric Representation of a Helix

— Helix:  $x(t)=r*\cos(t)$ ,  $y(t)=r*\sin(t)$ ,  $z(t)=ct$

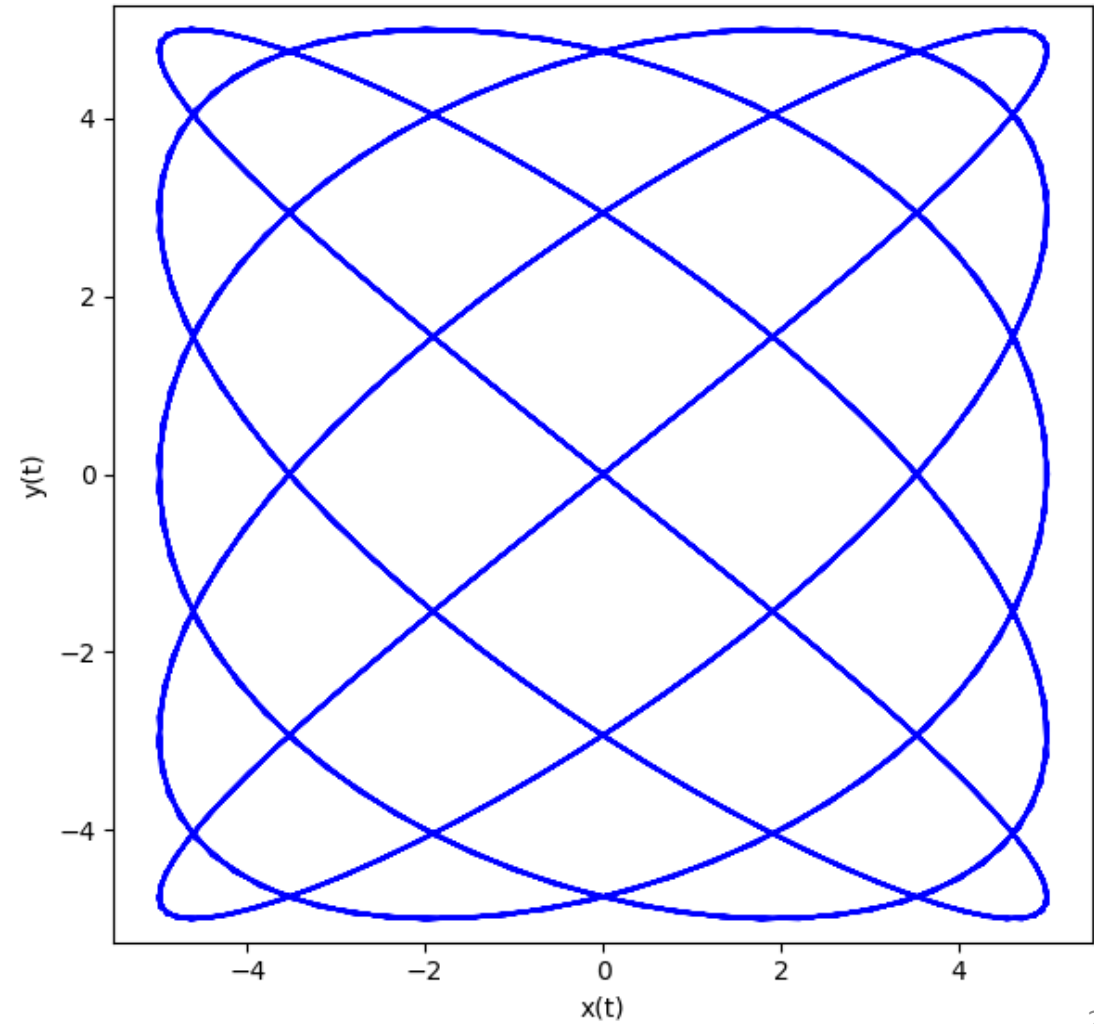


## Some Important Relations

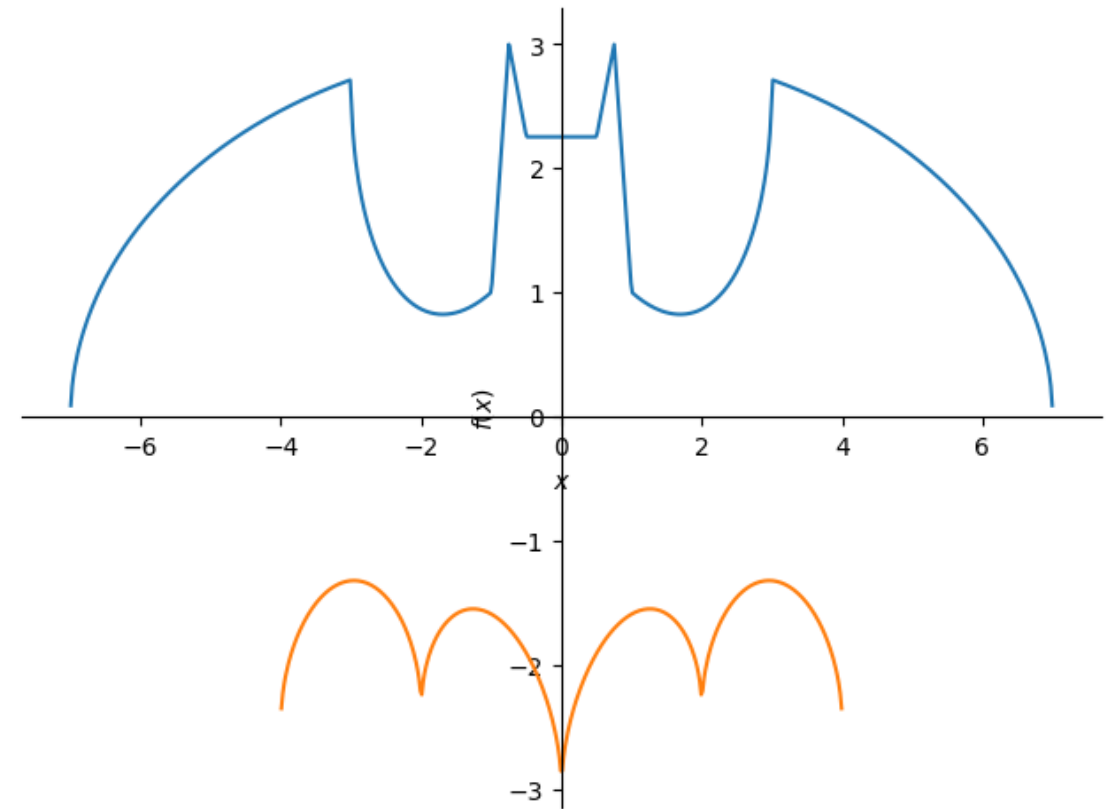
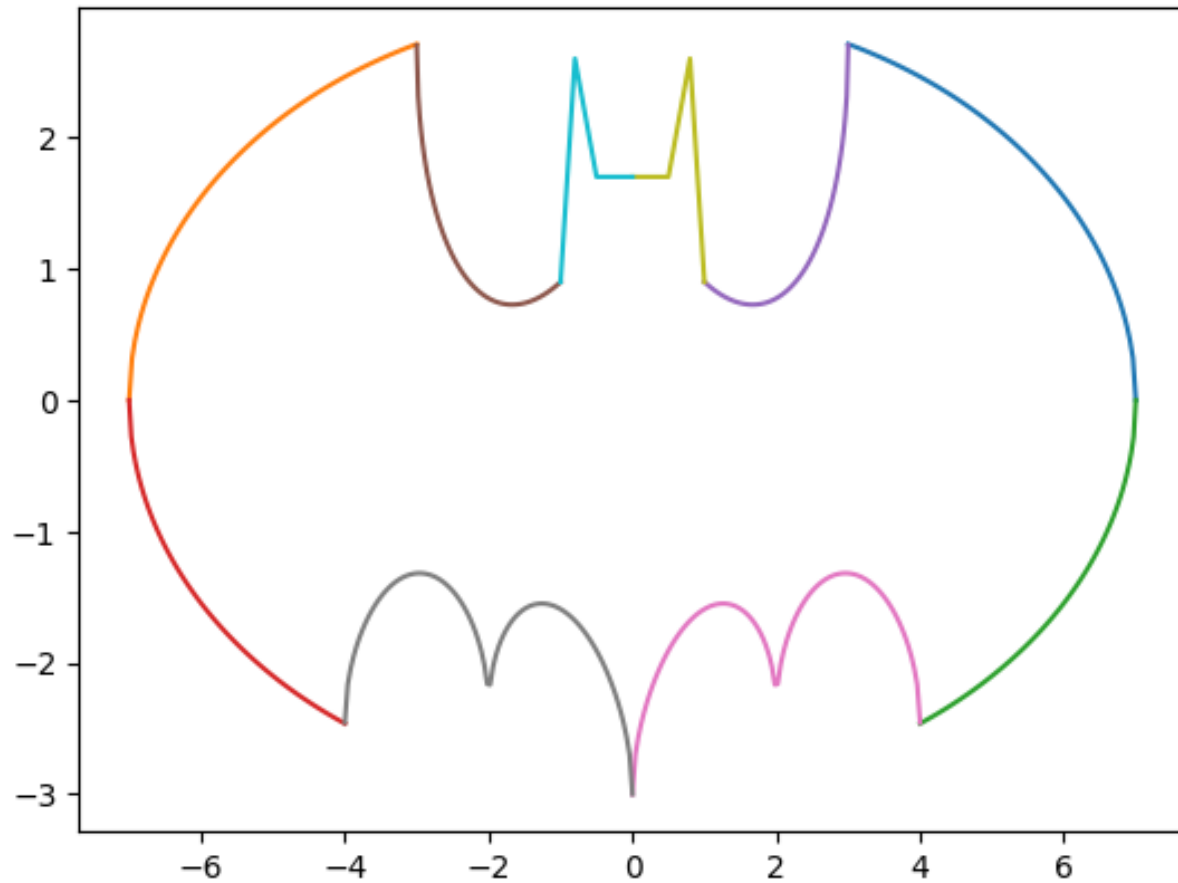
Butterfly Curve



Lissajous Curve



## Some Important Relations





## Formal definition of parametric curve

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Let  $f$  and  $g$  be the continuous functions on the interval  $I$ . The set of all the points  $(x, y) = (f(t), g(t))$  in the cartesian, as the  $t$  varies over  $I$ , is the graph of parametric equations  $x = f(t)$  and  $y = g(t)$ , where  $t$  is the parameter. A curve is a graph along with the parametric equations that define on it.

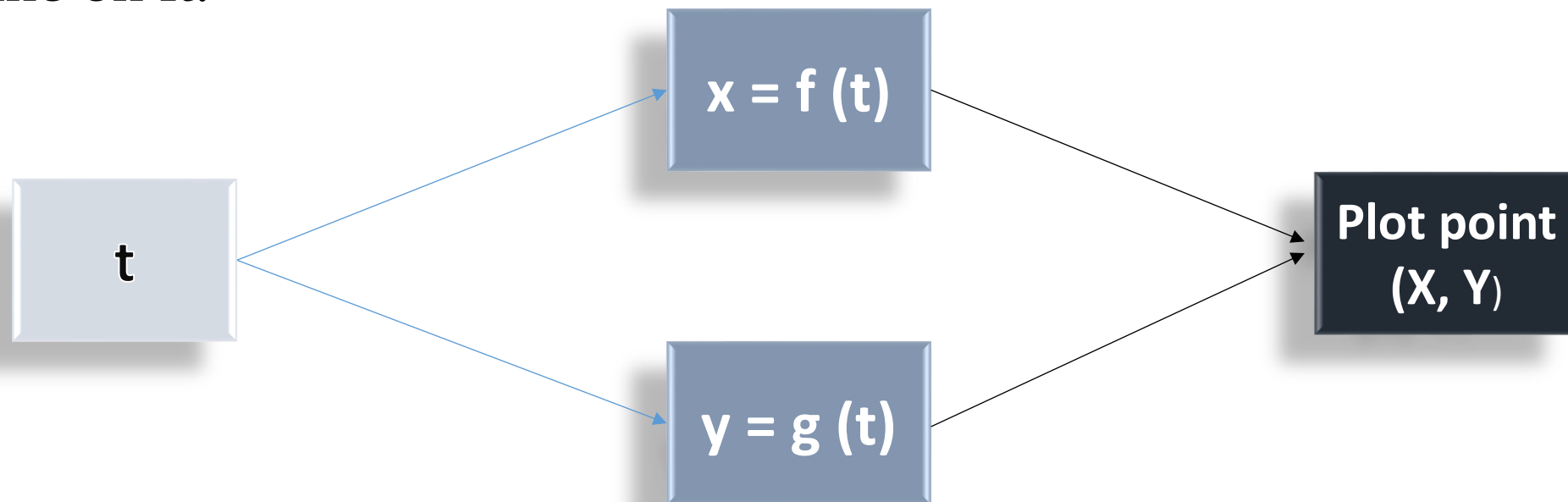
## Formal definition of parametric curve

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## Formal definition of Parametric Curves

Let  $f$  and  $g$  be the continuous functions on the interval  $I$ . The set of all the points  $(x, y) = (f(t), g(t))$  in the cartesian, as the  $t$  varies over  $I$ , is the graph of parametric equations  $x = f(t)$  and  $y = g(t)$ , where  $t$  is the parameter. A curve is a graph along with the parametric equations that define on it.



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# Practice Questions